



ACADEMIC MAJOR PROJECT PRESENTATION

AI Student Performance Prediction System

An Intelligent Machine Learning Approach for Early Academic Risk Identification

DEPARTMENT / COURSE

B.Tech CSE (AI / ML)

PRESENTED BY

Student Name

UNDER GUIDANCE OF

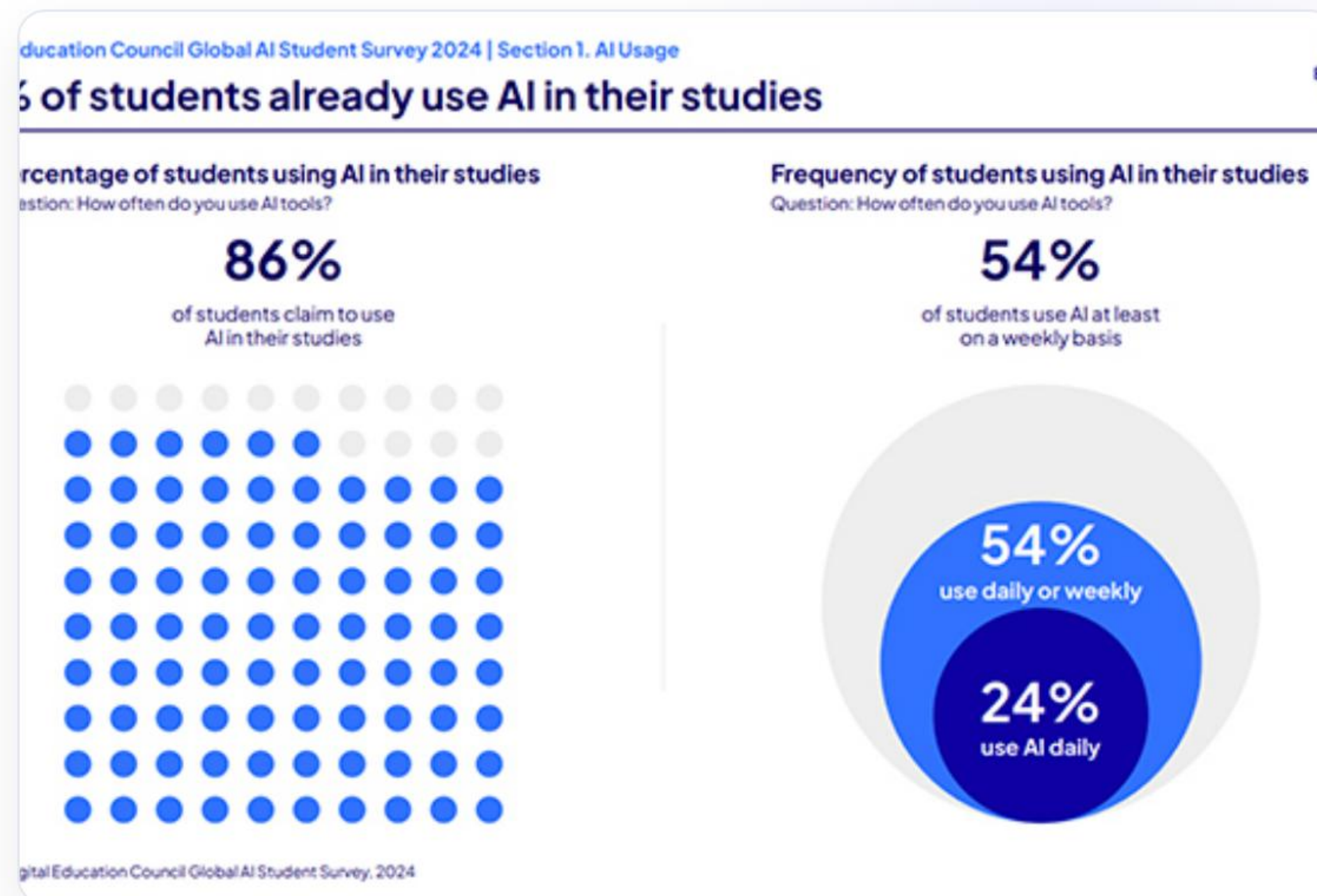
Guide Name (College Name)

i Introduction

OVERVIEW

Transforming Education with Predictive AI

-  **Student Performance Prediction:** Using data-driven models to forecast academic outcomes based on historical behavior and metrics.
-  **Role of AI in Education:** Automates pattern recognition, replacing subjective guesses with statistical accuracy.
-  **Strategic Importance:** Empowers institutions to proactively intervene before final examinations take place.



⚠ Problem Statement



Late Risk Detection

Faculty typically identify weak or struggling students only after mid-terms or final exams, leaving insufficient time for corrective remedial support.



Manual Workload

Evaluating multi-factorial academic records manually across hundreds of students is slow, tedious, and highly prone to human oversight or bias.



Lack of Predictive Tools

Educational institutions lack integrated intelligent systems capable of aggregating diverse student behavioral parameters into accurate forecasts.

🎯 Project Objectives



Predict Academic Performance

Develop machine learning algorithms to accurately compute and predict students' overall final marks and grades based on continuous tracking.



Identify At-Risk Students

Flag academically vulnerable students early in the semester to enable timely intervention, tutoring, and personal counseling.



Improve Pass Rates

Enhance total institutional pass percentages and average academic scores through data-backed academic management strategies.



Empower Faculty Decisions

Provide professors and mentors with visual insights and data dashboard summaries to make informed, objective pedagogical choices.

</> Technologies & Tools



Python Stack

Core programming language utilized for model creation, data manipulation, and modular machine learning logic.



Pandas & NumPy

High-performance data manipulation libraries used for feature extraction, cleaning, and matrix computation.



Scikit-learn

Comprehensive machine learning toolkit used for regression, classification algorithms, tuning, and evaluation.



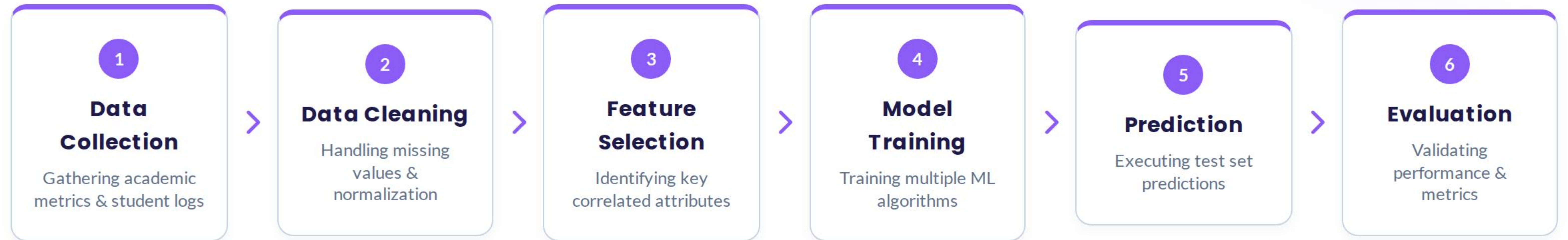
Matplotlib & Jupyter

Matplotlib for generating visual graphs and metrics; Jupyter Notebook for rapid prototyping and experiment tracking.

Dataset & Feature Architecture

Feature Name	Data Type	Value Range / Description	Impact Level
Attendance Rate	Numerical (%)	0% - 100% total lecture participation	HIGH
Study Hours / Week	Numerical (Hours)	1 - 45 weekly self-study duration	HIGH
Previous GPA / Marks	Numerical Score	Historical academic exam percentages	CRITICAL
Assignment Completion	Numerical (%)	Continuous assessment submission record	HIGH
Internal Assessment	Numerical Score	Mid-semester quiz & test marks	CRITICAL
Class Participation & Sleep	Categorical / Num	Engagement level & nightly sleep hours	MEDIUM

Machine Learning Workflow



Algorithms Evaluated

BASELINE MODELS

Comparative Machine Learning Models

-  **Linear Regression:** Provides a continuous baseline; efficient but struggles with complex non-linear trends.
-  **Decision Tree:** Captures hierarchical decision rules; simple to interpret but prone to overfitting.
-  **Support Vector Machine (SVM):** Effective in high-dimensional feature spaces with strong margin separation.

TOP PERFORMER

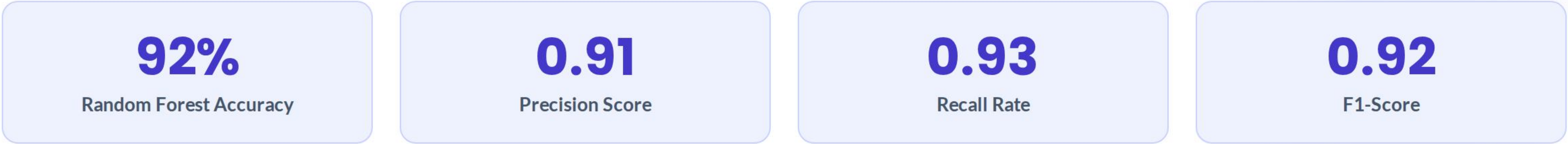
Random Forest Classifier

Ensemble learning technique combining multiple decision trees to boost accuracy and generalize seamlessly on unseen student datasets.

Key Advantage:

Achieved highest overall precision and lowest variance across all cross-validation folds.

Model Results & Performance



Algorithm Accuracy Comparison (%)



System Advantages

-  **Proactive Early Warning:** Automatically flags academically endangered students weeks before mid-term evaluations.
-  **Data-Driven Academic Planning:** Replaces intuition-based assessments with continuous quantitative tracking.
-  **Personalized Student Mentorship:** Helps tutors tailor specific study plans according to weak individual features (e.g., attendance vs. assignments).
-  **Institutional Retention Boost:** Directly decreases dropout rates and improves overall institutional success rates.

Future Enhancements



Deep Learning

Incorporating Recurrent Neural Networks (RNNs) and Transformers to evaluate longitudinal student learning curves.



Real-Time Dashboard

Building an interactive web dashboard for department heads and faculty to view real-time risk heatmaps.



Mobile Application

Creating a mobile interface for students to track their progress, receive risk notifications, and view target study hours.



AI Guidance Chatbot

Integrating a LLM-powered virtual assistant offering personalized study schedule advice and remedial recommendations.

PROJECT SUMMARY

The AI Student Performance Prediction System successfully demonstrates how machine learning (specifically Random Forest) can accurately forecast academic outcomes, enable proactive interventions, and elevate educational success.

Thank You!

Questions & Answers Session

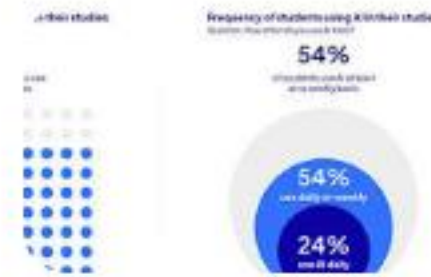


B.Tech CSE (AI / ML)



AI Academic Department

Image Sources



<https://campustechnology.com/-/media/EDU/CampusTechnology/2024/08/20240828DECsurvey.jpg>

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